

HW5

STAT-S103 Teaching Team

7/2/2025

The Effect of High Alcohol Consumption on Liver Scarring Among 21-50 Year Old NHANES Subjects

Background [10pts]

- Mention the rationale and objective your study.
- Include at least one reference to provide bullet points of relevant background information (copy/paste links is ok).

Methods [10pts]

- Mention the NHANES dataset, say something brief to provide information about it to readers
- Briefly describe your variables (look at the info. PDF)
- Mention your inclusion/exclusion criteria.
- Describe any manipulation of variables, such as:
 - creating categorical variables from continuous or discrete variables
 - or, combining categories to recode an existing categorical variable
- Mention the tests you used in your study's baseline characteristics table

(In a real study, you also mention having used R and what version, mention specific packages used, but it's fine not to include those for this exercise)

Results

Figure 1. Flowchart showing the inclusion/exclusion criteria. [5pts]

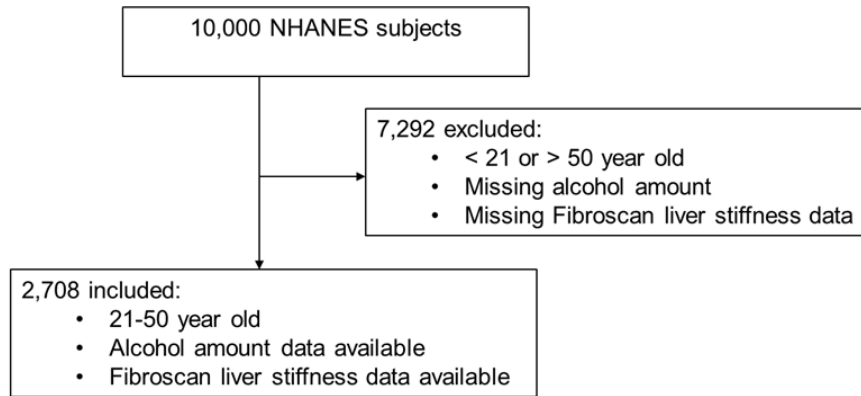


Figure 1. Flowchart of the study population.

Table 1 (baseline characteristics) [10 pts]

	All subjects (N=2708)	Outcome-No (n=2603)	Outcome-Yes (N=105)	p-value
High Alcohol	468 (17.3%)	432 (16.6%)	44 (41.9%)	< 0.0001
Age	34.8 (8.4)	34.1 (8.7)	37.4 (7.7)	< 0.0001
Sex - Female	1265 (46.7%)	1236 (47.5%)	29 (27.6%)	< 0.0001
Sex - Male	1443 (53.3%)	1367 (52.5%)	76 (72.4%)	

Table 2 (univariable model) [10 pts]

	OR [95% CI]	p-value
High Alcohol	3.62 [2.42 - 5.40]	0.0001

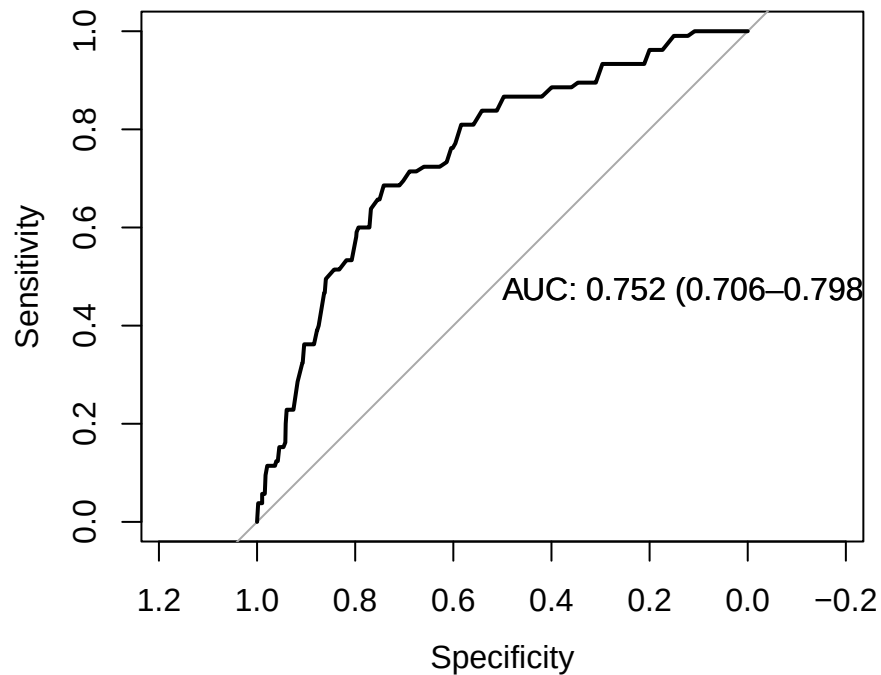
The odds of hypertension among the high alcohol consumption group (5 or more per day) is 1.58 times the odds among the reasonable alcohol consumption group.

Table 3 (multivariable model) [10 pts]

	OR [95% CI]	p-value
High Alcohol	3.56 [2.35 - 5.36]	< 0.0001
Age	1.07 [1.05 - 1.10]	< 0.0001
Sex (ref: Female)	2.02 [1.31 - 3.20]	0.0019

The odds of hypertension among the high alcohol consumption group (5 or more per day) is 1.58 times the odds among the reasonable alcohol consumption group, adjusting for age and sex.

Figure 2. ROC curve for the multivariable model [5 pts]



Conclusion/Discussion [10 pts]

Provide your interpretation of your study's results:

- Effect of the exposure - ORs, were they expected? Did you find a meaningful/interesting association?
- Crude vs. Adjusted OR – notable confounding detected?
- Any other confounding you did not account for that might have been important?
- Possible caveats and opportunities for improvement
- Possible biases?
- Sample size sufficient?
- Any public health message?
- Generalizability of your findings, if you think there could be any?

HW5.Rmd file upload and correctness of the code [15 pts]

Posting your results on Yellowdig [10 pts]

Commenting on three of your classmates' posting [extra credit: 5 pts]